

Sayantan Chakraborty

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Education

University of Dhaka, Dhaka, Bangladesh
B.S.S. in Health Economics

09/2023 – 11/2027 (Expected)

Work Experience

Undergraduate Research Assistant in Behavioral Economics

University of Pittsburgh, Pittsburgh, USA (Advisor - [LESTER R. LUSHER](#))

12/2024 – 06/2026

- Extracted and analyzed large-scale Google Trends data to support the replication of major economics papers.
- Developed Stata scripts to automate pairwise correlation analysis and efficiently manage missing data points.
- Leveraged language learning models to analyze linguistic patterns in paper titles across altscore categories.
- Curated and cross-referenced extensive datasets in Excel, ensuring high data accuracy and integrity.
- Developed scalable data preparation methods, coordinating with stakeholders to improve analysis workflows.

Skills

Programming & Software: Python, R, Stata, LaTeX, Visual Studio Code, KoboToolbox, Qualtrics.

Data & AI Capabilities: Statistical Analysis, Data Visualization, Prompt Engineering, Sentiment Analysis.

Economics & Research: Cost Analysis, Policy Evaluation, Academic Writing, Data Analysis, Data Management.

Professional Skills: Public Speaking, Leadership, Teamwork, Microsoft Office, Google Workspace.

Publications

- Roy, T., Bhattacharjee, U., Saha, S. K., **Chakraborty, S.**, Howlader, K., Bhattacharjee, T. (2026). [Toward Uncertainty Quantification in Modern Art](#). arXiv preprint arXiv:2608.04038, August 2026.
- Banik, S., **Chakraborty, S.**, Das Toma, B., Ghafoor, Z., Bhattacharjee, U., Howlader, K., Roy, T. (2026). [TRAPS: Therapeutic Response Analysis via Pathway-informed Stratification](#). arXiv preprint arXiv:2606.09898, June 2026.
- **Chakraborty, S.**, Mouno, M., Bhattacharjee, T., Roy, T., Bhattacharjee, U., Pretom, S. C., Roy, A. (2026). [Intelligent Health Intervention System for Syndemic Management Among University Students: An AI-Driven Approach Using Reinforcement Learning](#). Accepted for oral presentation at the AAAI 2026 Spring Symposium Series (accepted but not presented).
- Bhattacharjee, U., Roy, T., Howlader, K. C., **Chakraborty, S.**, Bhattacharjee, T., Roy, A. (2025). [Benchmarking Bioemulation-1: Establishing a Baseline for Protein Structural Ensemble Prediction](#). 2025 AIChE Annual Meeting, Boston, MA, November 2025.
- Bhattacharjee, U., Roy, T., Bhattacharjee, T., Saha, U., **Chakraborty, S.**, Roy, A. (2025). [EmissAI: LLM-Powered Analysis of Carbon Footprint](#). 2025 AIChE Annual Meeting, Boston, MA, November 2025.
- Ghafoor, Z., Islam, M. S., Howlader, K., Khondokar, M. R., Bhattacharjee, T., **Chakraborty, S.**, Roy, A., Bhattacharjee, U., Roy, T. (2025). [Enhancing Medical AI Safety Through Multi-Agent Evaluation and Iterative Refinement](#). 16th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB 2025), Philadelphia, PA, USA, October 2025.
- Roy, T., **Chakraborty, S.**, Bhattacharjee, U. (2025). [Unveiling Literary Depth: Leveraging LLaMA-Based NLP for Advanced Analysis of Popular Novels](#). Accepted for presentation at DULC 2025 Conference, University of Dhaka, Bangladesh.
- Bhattacharjee, U., Roy, T., Pretom, S. C., Kaium, A., **Chakraborty, S.**, Roy, A. (2025). [LlamaHealth: Assessing LLaMA's Performance in Multi-Class Healthcare Test Result Prediction](#). 8th International Conference on Engineering Research, Innovation and Education (ICERIE 2025), April 2025.
- Bhattacharjee, U., Roy, T., **Chakraborty, S.**, Roy, A. (2025). [George Washington Carver "I Can" Poster Challenge](#). Best-In-Show award at the George Washington Carver "I Can" Poster Challenge, Ames, Iowa, USA, February 2025.
- **Chakraborty, S.**, Bhattacharjee, U., Roy, T., Roy, A. (2025). [Evaluating the Impact of Vaccination Programs on Rubella Incidence: A Statistical Approach](#). RURS 1st International Student Conference and Research Fair 2024, University of Rajshahi, Bangladesh.
- Bhattacharjee, U., **Chakraborty, S.**, Roy, T. (2024). [AI-Driven Food Price Inflation Prediction and Country Clustering via LLMs](#). Norman Borlaug Poster Competition, Iowa State University, October 2024.